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25 January 1980
SUPERSEDING
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6 February 1978

MILITARY SPECIFICATION

CAP, UTILITY: CAMOUFLAGE

This specification is approved for use by U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for camouflage utility caps.

1.2 Classification. The caps shall be of the following types and sizes as specified (see 6.2):

Type I - With Marine Corps insignia
Type II - Without insignia

Schedule of Sizes

Extra Small
Small
Medium
Large
Extra Large

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMA-1), Headquarters, U. S. Marine Corps, Washington, D. C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8405

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V-T-276	- Thread, Cotton
UU-P-31	- Paper; General Specifications and Methods of Testing
UU-P-268	- Paper, Kraft, Wrapping
DDD-L-20	- Label; For Clothing, Equipage, and Tentage (General Use)

MILITARY

MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-E-20652	- Eyelets, Metallic and Eyelet Washers, Metallic, General Specification For
MIL-C-43468	- Cloth, Camouflage Pattern, Wind Resistant, Poplin, Cotton
MIL-T-43548	- Thread, Polyester, Cotton-covered
MIL-C-43594	- Cloth, Interlining, Polyester

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

LAWS AND REGULATIONS

U. S. Postal Service Manual

(Copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

Department of Defense (DOD) Standard Color Card of Official Standardized Shades for Sewing Threads 1968

(Copies may be obtained from the Color Association of the United States, Inc., 200 Madison Avenue, New York, New York 10016.)

3. REQUIREMENT

3.1 Standard sample. Standard samples are furnished solely for guidance and information to the contractor (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 First article. Unless otherwise specified (see 6.2) before production is commenced, a finished cap shall be submitted, or made available to the contracting officer or his authorized representative for approval. The approval of the first article authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all provisions of this specification. The first article shall be manufactured by the contractor at the same facilities to be used for the manufacture of the production item.

3.3 Materials.

3.3.1 Basic fabric. The basic material for the caps, including the sweatband shall be cloth, camouflage pattern, wind resistant, poplin, cotton, conforming to type I of MIL-C-43468.

3.3.2 Interlining. The cloth for interlining the crown and outside band shall be cloth, interlining, polyester confirming to MIL-C-43594.

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3.3.3 Thread. The cotton thread shall conform to V-T-276, soft finish, type 1A3, ticket No 40, 3 ply for all seaming and stitching or as an alternate, cotton-covered polyester thread conforming to ticket No. 50, 2 or 3 ply of MIL-T-43548, may be used.

3.3.3.1 Color and colorfastness. The thread shall be dyed olive drab shade S-1, Cable No. 66022. (See DOD Standard Color Card of Official Standardized Shades for Sewing Threads, 1968). The thread shall show fastness to perspiration, combined laundering and bleaching, and light equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show good fastness to perspiration, combined laundering and bleaching and light.

3.3.4 Inner visor. The inner visor shall be made out of a nonwoven material produced from a highly purified alpha cellulos vegetable fiber, wet web impregnated with approximately 22 + 5 percent compounded neoprene and shall conform to the following physical requirements, when tested as specified in Section 4, Table II.

Requirement	Machine Direction	Crosswise Direction
Stiffness (load lb., min)		
Original	2.4	1.5
After laundering	2.4	1.5
After wet (dry) cleaning	2.2	1.4
Tensile strength (lb per in, min)	156	103
Stretch (percent max)	11	22
Tearing strength (gms min)	1770	2040
Weight (lb per sq yd) 2.21 + 5 percent		
Thickness 0.075 $\pm$ 0.003 inch		

3.3.4.1 Water absorption. The material shall not absorb more than 30 percent water when tested as specified in Section 4, Table II.

3.3.4.2 Resistance to perspiration. The material shall show no embrittlement, development of stickiness, or delamination when subjected to the perspiration test specified in Section 4, Table II.

3.3.4.3 Accelerated aging. The material shall show no embrittlement, development of stickiness, or delamination after 15 days in an oven with circulating air at 100°C when tested as specified in Section 4, Table II

3.3.4.4 Resistance to water. The material shall not shrink or warp after soaking in water 24 hours when tested as specified in Section 4, Table II.

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3.3.4.5 Directional marking. The finished alpha cellulose material shall be marked by directional symbols to designate the greater flexibility of the material. The marking shall be legibly and clearly designated on the textured side of the material.

3.3.5 Eyelets The eyelets shall be brass, rolled flange type, with the barrel scored 6 to 8 scores. The eyelets shall conform to MIL-E-20652, sheet 1A-BE108 and be enameled a green shade to closely match the dark green shade of the basic fabric.

3.3.6 Labels. Each cap shall have an identification label and a size label as follows. The labels shall have colorfastness to laundering as specified in DDD-L-20.

3.3.6.1 Identification label. The identification label shall conform to type I or IV, class 1 of DDD-L-20. The content and form shall be as follows:

Nomenclature (Item description)
Contract Number
Name of Contractor

3.3.6.2 Size label. The size label shall conform to type I or IV, class 2 of DDD-L-20. The label shall show the size designation specified in 1.2 and the stock number as shown by the following example:

$$\frac{\text{Small}}{\text{Stock Number}}$$

3.3.6.3 Combination identification and size label. The identification and size label may be combined and the label shall be not less than 1 inch by 1-3/4 inches. The combination identification and size label shall conform to type I or IV, class 4 of DDD-L-20.

3.3.7 Insignia (type I cap). The Marine Corps insignia for the front of the type I cap shall conform to the design and dimensions shown on Figures 1 and 2. The insignia shall be applied by decalcomania marking, transferred by heat and pressure. The transfer shall produce an imprint on the basic material which is clear and distinct with sharp outlines, the filled-in areas a solid black and the open areas of rope and globe clearly visible. The transferred marking shal be clearly visible after being subjected to three launderings when tested as specified in Section 4, Table II.

3.4 Design. The cap shall be the design shown on Figures 1 through 4, having a lined two-piece crown shaped with eight evenly spaced folded and stitched darts; two metal eyelets at each side of the crown for ventilation; a lined outside band and a sweatband made from one ply of basic material; and a flexible, quilted hinge type visor.

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3.5 Patterns. The standard patterns furnished by the Government shall not be altered in any manner and shall be used only as a guide for cutting working patterns. The patterns show size, and are marked or notched for proper cutting, placement, and assembly of component parts.

3.5.1 Pattern parts. Caps shall be cut from materials specified herein in accordance with the pattern parts indicated below:

Item	Pattern Nomenclature	No. of cut parts
	Material: Cloth, Camouflage	
1.	Crown	2
2.	Top visor cover	1
3.	Under visor cover	1
4.	Outside band	1
5.	Sweatband	1
	Material: Cloth, Interlining	
6.	Crown	1 1/
7.	Outside band	1
	Material: Alpha Cellulose Fiber	
8.	Inner visor	1
	Marker	
9.	Crown	

1/ Crown interlining may be cut 1/8 inch smaller than crown all around entire periphery.

3.6 Construction. The caps shall be manufactured in accordance with the operation requirements specified in Table I. The manufacturer is not required to follow the exact sequence of operations as listed.

3.6.1 Stitches, seams and stitchings. The stitch, seam and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more stitch or seam types are specified for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (under thread) shall be on the inside of the cap. The number of stitches per inch as specified in Table I indicates the minimum and maximum number permitted.

3.6.1.1 Tacking and backstitching. Ends of seams and stitching, produced with 301 type stitch, when not caught in other seams or stitching shall be backstitched not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back of break not less than 1/2 inch. When stitch type 401 is used, ends of seams and stitching shall be caught in other seams of stitching, and skipped stitches and thread breaks may be repaired using stitch type 301 backstitched not less than 1/2 inch on each side of the skip or break.

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3.6.2 Shade marking. The component parts shall be marked to insure a uniform shade pattern and size throughout the cap. Any method of shade marking may be used except:

(1) Corrosive metal devices.
(2) Adhesive-type tickets which show discoloration and the adhesive mass which adheres to the material upon removal of the tickets.

TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD	
					NDL	BOBV LPR
1.	Cutting. a. Cut cap in strict accordance with patterns furnished by the Government. b. Cut parts in the direction of warp as indicated by directional lines on pattern. c. Cut component parts of cap from one piece of basic material, except sweatband and underply crown, which may be cut from ends. Cloth shall be cut with the face of the cloth to finish on outside of cap.					

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOBV LPR
1.	Cutting. (cont'd) d. Cut inner visor with pattern placed side to side across the width (crosswise direction) of material specified in 3.3.4, so that the greater flexibility of the cut visor is from side to side (see Figure 4). e. Cut interlining for crown and outside band in accordance with patterns. The interlining for the outside band shall be cut 1/4 inch narrower than the cloth.					
2.	Replacement of damaged parts. a. Replace at time of cutting any parts of the cap containing holes, seriously weakening defects such as smashes, mutiple floats, loose or weakening slubs, etc. expected to develop into holes, shade bars and dye streaks or unsightly slubs, etc. affecting appearance. b. Replace any part damaged during manufacturing process by needle chews which may develop into holes, sissors or knife cuts, tears, holes, mends, burns and large drill holes in the material affecting appearance.					
3.	Shade marking. a. Mark, ticket or bundle all component parts (see 3.6.2) cut from one piece of basic material, except those parts cut from ends, (see 1.c) to insure uniform shade and size throughout the cap. Any size part cut from ends shall be identified as to size.					

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOB/ LPR
4.	Label(s)/marking caps.					
	a. The identification and size label(s) as specified in 3.3.6 shall be legibly applied and correctly positioned on sweatband without smearing. Marking shall be thoroughly dry prior to assembly of cap.					
	b. Labels shall be applied to sweatband either to the left or right side of the joining seam.					
5.	Make crown.					
	Finished appearance. Darts shall finish without gathers, puckers or pleats.					
	a. Position the crown interlining on inside ply of crown with all edges even and stitch together across full width with single rows of stitches, perpendicular to each other forming an X at the center. Position the top ply of the crown on top of the interlining with edges even and seam 1/8 inch from edge.	301	SSa-1	10-14	40/3	40/3
	b. Mark position of darts and eyelets in accordance with the pattern marker.					
	c. Fold and stitch crown darts in accordance with marks, tapering upper end of darts to a point. The point shall be securely tacked or backstitched.	301	OSf-1	10-14	40/3	40/3
	d. Fold crown at upper edge of three front panels formed by darts and stitch 1/16 inch from folded edge.	301	OSF-1	10-14	40/3	40/3

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOB/ LPR
5.	Make crown. (cont'd)					
	e. Apply Marine Corps insignia (type I cap) to center front of crown with heat and pressure. The imprint shall be clear and distinct and shall be vertically centered on front of crown 1/4 ± 1/8 inch below stitched upper edge of center front panel, or apply Marine Corps insignia to center front panel with heat and pressure prior to making the darts.					
6.	Attach eyelet to crown.					
	Finished appearance. Eyelet shall be positioned and securely attached through each side of crown and securely clinched on inside without distorting and cutting material or splitting the flange of the eyelet. In the finished cap, the eyelets shall be centered 1 ± 1/8 inch above top edge of outside band.					
	a. Attach two metal eyelets through each side of crown. Eyelets shall be centered 1-1/4 ± 1/8 inches apart and equidistant from back and front side darts.					
7.	Stitch darts around crown.					
	a. Stitch around crown 1/8 to 1/4 inch from lower edge. When stitching, turn the fold of all darts, except the two at the back, towards the front of crown. Turn the fold of the two back darts toward the center back of crown.	101 or 301 or 401	SSa-1	6-8	40/3	40/3

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOB/ LPR
8.	Make outside band.					
	a. Position the interlining on the underside of the outside band with top edges even and stitch together 1/8 inch from the edge for the entire length of the band.	101 or 301	SSa-1	6-8	40/3	40/3
			SSa-1	10-14	40/3	40/3
	b. Join ends of band with a 1/4 inch seam and securely backstitch lower ends of stitching.	301	SSa-1	10-14	40/3	40/3
	c. Press seam open or manually spread open seam.					
	d. Fold under lower edge of band 1/4 to 5/16 inch and stitch 1/8 inch from folded edge catching the interlining in the stitching.	301 or 401	EFa-1	10-14	40/3	40/3
			EFa-1	10-14	40/3	40/3
	-or-					
	If operation 8d is preformed before oper 8b, the ends of the seam allowance shall be trimmed on a diagonal to reduce bulkiness and eliminate exposure of the interlining when the sweatband and outside band are joined together (see oper 11g).					
9.	Make sweatband.					
	a. Join ends of band with a 1/4-inch seam and securely backstitch lower ends of the stitching.	301	SSa-1	10-14	40/3	40/3
	b. Press seam open or manually spread open seam.					
	c. Fold under lower edge of sweatband 1/4 inch and stitch 1/8 inch from folded edge.	301 or 401	EFa-1	10-14	40/3	40/3
			EFa-1	10-14	40/3	40/3

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOB/ LPR
10.	Make visor.					
	Finished appearance. The visor shall be smooth and free of wrinkles, gathers, puckers, or other distortions. Quilting of visor shall be uniform.					
	a. Position top and bottom plies (basic material) for visor face to face with outer edge even and notches matching. Stitch together 1/4 inch from the raw edge, distributing the fullness of bottom ply between notches.	301 or 401	SSa-1	10-14	40/3	40/3
		401	SSa-1	10-14	40/3	40/3
	b. Place seamed pieces of basic material on inner visor with top piece next to the inner visor and outer edge of all pieces aligned.					
	c. Join together all pieces of visor around outer edge, stitched 1/4 to 5/16 inch from edge.	301 or 401	SSa-1	6-8	40/3	40/3
		401	SSa-1	6-8	40/3	40/3
	d. Turn bottom piece to underside of inner visor forming a smooth front edge with an even binding affect. Raise stitch the binding affect through all plies with the raise stitching 1/16 to 1/8 inch on the seamed edge of simulated binding.	301	-	6-8	40/3	40/3
	e. Quilt visor through all plies with 4 rows of stitches evenly spaced 3/8 to 1/2 inch apart and parallel to outer edge of visor.	301	SSv-4	6-8	40/3	40/3
11.	Assemble cap.					
	Finished appearance. The outside band and sweatband shall be smoothly and evenly seamed to the cap without wrinkles, puckers, gathers, twists or other distortions. Outside band shall finish 1-1/4 $\pm$ 1/8 inches wide. Sweatband shall finish 1-3/8 $\pm$ 1/8 inch wide.					

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD NDL	THREAD BOBV LPR
11.	Assemble cap. (cont'd).					
	a. Place lower edge of crown between top edge of outside band and sweatband with all raw edges even and with joining seam of outside band aligned with joining seam of sweatband at center back of crown.					
	b. Join outside band and sweatband to crown 3/8 inch from lower edge of crown, catching one edge of cloth identification-size label in seam near inside center back of cap either to the left or right of joining seam of sweatband.	301 or 401	SSa-1	8-10	40/3	40/3
		401	SSa-1	8-10	40/3	40/3
	c. Turn down outside band and raise stitch 1/16 inch from seam.	301	LSq-2 (shows finished seam)	10-14	40/3	40/3
	d. Position sweatband on underside of visor in accordance with notches shown on pattern. Hemmed lower edge of sweatband shall be lapped on hinge of visor and abutted to the back edge of inner visor.					
	e. Seam sweatband to visor not more than 1/16 inch from hemmed lower edge, stitched through hinge of visor but not through inner visor.	301 or 401	LSb-1	10-14	40/3	40/3
		401	LSb-1	10-14	40/3	40/3
	f. Seam outside band to sweatband 1/16 inch from hemmed lower edges and with hemmed lower edge of outside band lapped on hinge of visor and aligned with back edge of inner visor.	301	SSc-1	10-14	40/3	40/3
	g. Seam outside band to visor 1/16 inch from hemmed lower edge, stitched through hinge of visor but not through inner visor. From that point continue to stitch lower edge of outside band and sweatband together 1/16 inch from the hemmed edge.	301	LSb-1	10-14	40/3	40/3

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TABLE 1 MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	THREAD	
					NDL	BOB/ LPR
12.	Clean cap. a. Trim all ends of stitching and remove all loose ends of thread from cap. Remove all spots and stains, and remove all shade tickets and marking without injury to the material.					

3.7 Finished measurements. The finished measurements of the caps shall conform to the following:

Size	Head measurement (inches) 1/
Extra Small	21-1/4
Small	22
Medium	22-3/4
Large	23-1/2
Extra Large	24-1/4

3.8 Workmanship. The finished caps shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

1/ Tolerance: + 1/4 inch, - 0

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-105.

4.4 First article inspections. When first article is required, inspection shall be made of a completely fabricated cap for all provisions of this specification.

4.5 Quality conformance inspection.

4.5.1 Testing of components. Testing of components shall be in accordance with applicable subsidiary specifications and standards to the extent applicable, except that the specification shall govern in the event of conflict. The method of testing specified in FED-STD-191 and UU-P-31, wherever applicable, and as listed in Table II shall be followed. All requirements are applicable to sample unit. When the data as to number of determinations per sample unit and the results reported as columns are not specified in Table II, they shall be as required by the referenced test method. The unit for expressing lot sizes and sample unit for testing each component shall be in accordance with applicable subsidiary specification and as follows:

Component	Lot size expressed as	Sample unit for testing
Nonwoven fibrous material (inner visor)	yards	1 square yard
Eyelets	each	15
Insignia (decalcomania)	each	15

4.5.1.1 Sampling for testing. Government furnished materials shall not be sampled or tested unless specifically requested by the Government. Unless a component is covered by a subsidiary specification that specifies the inspection level and AQL for testing, the inspection level shall be S-1 and the AQL shall be 4.0 of MIL-STD-105.

4.5.1.2 In-process inspection. Inspection shall be performed during the manufacturing of the end item to ascertain that the operations are performed as specified herein. The Government reserves the right to determine the acceptability of construction methods not performed in accordance with specified requirements for operations or subassemblies.

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Table II. Test Methods

Components	Characteristics	Rqmt. para.	Test method	No. Determinations per sample unit	Results reported as
Inner visor	Material identification	3.3.4	1/	1	Pass or fail
	Compounded neoprene content	3.3.4	1/	1	To the nearest percent
	Stiffness:				
	Original	3.3.4	5202 2/		
	After laundering	3.3.4	5610/5202 2/		
	After wet (dry) cleaning	3.3.4	5622/5202 2/		
	Tensile strength	3.3.4	171 3/	10	To the nearest 1.0 lb.
	Stretch	3.3.4	171 3/	10	To the nearest percent.
	Tearing strength	3.3.4	170 3/	5	To the nearest Gm.
	Weight	3.3.4	5041 2/	1	To the nearest lb/sq yd.
	Thickness	3.3.4	5030 2/	1	To the nearest 0.001 inch
	Water absorption	3.3.4.1	5502 2/	1	Pass or fail
	Perspiration	3.3.4.2	5680 2/	1	Pass or fail
	Accelerated aging	3.3.4.3	4/	1	Pass or fail
	Shrinkage	3.3.4.4	5/	1	Pass or fail

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Table II. Test Methods

Components	Characteristics	Rqmt. para.	Test method	No. Determinations per sample unit	Results reported as
Insignia (applied on swatches of basic material)	Insignia shall be clearly visible after laundering 6/	3.3.7	5610	3	Pass or fail

1/ A certificate of compliance is required and will be acceptable for the requirement stated. (see 4.1.1)

2/ FED-STD-191.

3/ UU-P-31.

4/ The material shall be kept in a forced circulating air over 15 days at 100°C.

5/ The material shall be soaked in water at room temperature for 24 hours.

6/ Test to be performed three times prior to evaluation.

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4.5.2 Examination of the end item. The end item shall be examined for defects in 4.5.2.1 and 4.5.2.2. The sample unit for the examination shall be one cap.

4.5.2.1 Visual examination. The end item shall be visually examined for defects. The inspection level for this visual examination shall be level II of MIL-STD-105. The AQL for this examination shall be 2.5 defects per 100 units (DHU) for major defects, 10.0 DHU for major and minor A (combined) defects, and 15.0 DHU for total (major and minor A and B combined) defects. Defects shall be classified as follows:

Examine	Defect	Classification		
		Major	Minor	
			A	B
Material defects and damages	a. Any hole.	X		
	b. Any weakening defects such as a smash, multiple float, loose slub which may develop into a hole.	X		
	c. Shade bar or unsightly slub, affecting appearance.	X		
	d. Needle chew, cut, tear or mend.	X		
	NOTE: Material defects and workmanship damages are to be classified as indicated above only when the condition is one which weakens the fabric or garment, or when it is so conspicuously located as to be clearly noticeable. Nonweakening conditions which are inconspicuous shall not be classified as defects.			
Shaded parts	Any outside part shaded (affecting appearance).		X	
Cleanness	a. Spots or stains of an obvious permanent nature.		X	
	b. Removable spots or stains, clearly noticeable.		X	

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	c. Thread ends not trimmed or loose threads not removed.			X
	d. One or more shade tickets not removed.			X
	e. Shade stamp marking exposed on outside of caps, seriously affecting appearance.	X		
Component and assembly	a. Any defective component or defect of assembly (unless otherwise classified).		X	
	b. Any component part or required operation omitted (unless otherwise classified).	X		
Seam and stitching	a. Accuracy of seaming:			
	- Part of cap caught in any unrelated operation of stitching.		X	
	- Seams twisted, puckered, or pleated (unless otherwise classified herein).		X	
	- End(s) of stitching when not caught in other seams of stitching backstitched less than 1/4 inch.		X	
	- Stitching overlapping the end(s) of thread break less than 1/2 inch.			X
	b. Gage of stitching (top, edge, raised or quilting stitches):			
	- Irregular, i.e, unevenly gaged or various stitching not uniform, affecting appearance (to be scored when conditions exists on major portion of seam).		X	

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	- Edge of raised stitching sewn too close to edge, resulting in damage to cloth.	X		
	- Beyond range or width specified or varies more than 1/16 inch when no range is specified.			X
	c. Open seams:			
	- 1/8 inch or more but not more than 1/4 inch.			X
	- more than 1/4 inch but not more than 3/4 inch.		X	
	- more than 3/4 inch.	X		
	NOTE: One or more broken or two or more continuous skipped or run-off stitches constitute an open seam.			
	d. Raw edges:			
	- 1/4 inch or less.			X
	- more than 1/4 inch but not more than 3/4 inch.		X	
	- more than 3/4 inch.	X		
	NOTE: Raw edges not securely caught in stitching shall be classified as an open seam.			
	e. Run-off:			
	- Joining seams - When resulting in an open seam use "Open seam" classification.			
	- Top, edge, quilting or raised stitching (when not resulting in an open seam).			
	- On outside:			
	- more than 1/2 inch.		X	
	- 1/4 inch or more but not more than 1/2 inch.			X

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	f. Seams and stitch types:			
	- Not specified seam or stitch type.		X	
	- Looper thread of 401 stitch type on outside of finished cap.	X		
	g. Stitch tension:			
	- Loose tension on raised or edge stitching, i.e., loosely exposed loops of lower or top thread.		X	
	- Loose tension resulting in a loose seam.	X		
	- Tight tension (stitches break when normal strain is applied to seam or stitching).	X		
	NOTE: Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting normal pull in lengthwise direction of seam.			
	h. Stitches per inch (to be scored only when the condition exists on major portion of the seam or stitching), on all seams or stitching.			
	- less than minimum number specified by:			
	- not more than 2 stitches.			X
	- more than 2 but not more than 4 stitches.		X	
	- more than 4 stitches.	X		
	- one or more stitches in excess of maximum number specified.			X

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	i. Stitches skipped or broken:			
	- On edge or raised stitching when seam is seamed, turned and stitched:			
	- more than 1/4 inch.			X
	- On visor quilting stitches:			
	- more than 1/4 inch on one row of stitching.			X
	- more than 1/4 inch on two or more rows of stitching.		X	
Eyelets	a. Any omitted or added.		X	
	b. Defective eyelets (but securely clinched):			
	- 1 eyelet.			X
	- 2 or more eyelets.		X	
	c. Not properly clinched, i.e., loose or raw edges of material exposed around edge of eyelet on outside of cap:			
	- 1 eyelet.		X	
	- 2 or more eyelets.	X		
	d. Not specified type or size.		X	
	e. Position less than 7/8 inch or more than 1-1/8 inches from center of eyelet to top edge of outside band.			X
	f. Color and finish.			
	- Does not compare favorably with standard or approved sample.		X	
	- Surfaces not smooth.			X
	- Discoloration or stain clearly noticeable.			X
	- Abraded, scuffed, scratched or otherwise damaged.			X

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	g. Distance between eyelets from center to center less than 1-1/8 inches or more than 1-3/8 inches.			X
	h. Center of eyelets not equidistant from front and side darts by more than 1/4 inch.			X
Visor	a. Upper or lower plies of material badly wrinkled or puckered, pleated or excesively full between rows of quilting stitches.		X	
	b. Visor interlining caught in stitching joining bands to visor (to be scored when the condition exists on major portion of seaming).		X	
	c. Number of rows of quilting stitches (including row at finished edge of visor):			
	- 5 or less.		X	
	- 7 or more.			X

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	d. Visor not centered with center front of cap:			
	- more than 1/2 inch.	X		
	- more than 1/4 inch but not more than 1/2 inch.		X	
	e. Edge of outer material not firmly folded over edge of inner visor, i.e., excess material extends 1/8 inch or more beyond edge of inner visor.			X
	f. Finished edge of visor less than 3/16 inch or more than 5/16 inch wide.			X
	g. Edge of visor poorly shaped, affecting appearance.		X	
	h. End(s) of visor insecurely caught in stitching of outer and inner bands.	X		
	i. Width of visor at center less than 2-1/2 inches or more than 2-3/4 inches.			X
Band (outside)	a. Width less than 1-1/8 inches or more than 1-3/8 inches.			X
	b. Twisted, pleated or excessively full.	X		
	c. Irregular in width by 3/16 inch or more.	X		
	d. Back seam of band offset from mid-point between back crown darts by more than 1/4 inch.			X
Crown	a. Edge stitching along top edge at front omitted.	X		

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	b. End(s) of stitching along top edge at front:			
	- extending beyond outer dart(s) at either end by 1/2 inch or more.			X
	- not extending to outer dart(s) at either end by 1/4 inch or more.			X
	c. Height of crown at center front including band measures less than 3-3/8 inches or more than 3-5/8 inches.			X
	d. Height of crown at end of visor including band measures less than 2-7/8 inches or more than 3-1/8 inches.			X
	e. Crown puckered or pleated at joining to crown.		X	
Darts	a. Any dart omitted.	X		
	b. Folding of darts:			
	- back of darts not folded toward back of crown.		X	
	- center front darts not folded toward front of crown.		X	
	- all other darts on front and sides not folded toward front of crown.		X	
	c. Ends not properly tapered on one or more darts.			X
Insignia (Type I cap only)	a. Not specified type, size or color.		X	
	b. Not properly transferred, i.e., indistinct or blurred.		X	

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	c. Set on crookedly, affecting appearance.		X	
	d. Top edge placed less than 1/8 inch or more than 3/8 inch below folded (stitched) edged of crown.			X
	e. Off set between front darts by 3/8 inch or more.		X	
Sweatband	a. Ends of stitching joining outer band and sweatband at center back overlapping less than 1 inch.			X
	b. Sweatband exposed below bottom edge of cap by 1/8 inch or more:			
	- for a distance 1/2 inch or more, but not more than 1 inch.			X
	- for a distance of more than 1 inch.		X	
	c. Twisted or tight, affecting appearance of outside band.			X
	d. Joining seam of sweatband out of alignment with joining seam of outside band by more than 1/2 inch.			X
	e. Width less than 1-1/4 inches or more than 1-1/2 inches.			X
Labels	a. Size label omitted, incorrect or illegible.	X		
	b. Contractor's label:			
	- omitted, incorrect or illegible.		X	

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Examine	Defect	Classification		
		Major	Minor	
			A	B
	- Not securely caught in stitching when cloth label is used.		X	

4.5.2.2 Examination of finished measurements. The end item shall be examined for conformance to the finished measurements specified herein (see 3.7) The inspection level for this examination of the finished measurements shall be S-3 of MIL-STD-105. The AQL for this examination shall be 4.0 DHU for defects (one class). Any finished measurement deviating from the finished measurements specified herein shall be classified as a finished measurement defect.

4.5.3 Examination of preparation for delivery. Shipping containers fully prepared for delivery shall be examined for the defects listed below to determine conformance to packaging, packing and marking requirements. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 of MIL-STD-105. The AQL shall be 2.5 DHU.

Examine	Defect
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, improper taping, loose strapping, inadequate stapling, bulging or distortion of container.
Content	Number of bundles per shipping container is more or less than required. Number of caps per bundle is more or less than required. 1/

1/ For this defect, one bundle shall be examined from each shipping container in the sample.

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5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Ten (10) caps of one type and size only shall be tested together in a bundle, with visor lying flat, and cross-tied with cotton tape or twine.

5.1.2 Commercial. Commercial packaging shall afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The contractor may use his standard practice provided it meets the requirements specified for commercial packing.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Method 1. Two hundred (200) caps of one type and size only, packaged as specified in 5.2, shall be packed in a fiberboard shipping container with liner conforming to class weather resistant, size No. 3A of MIL-B-17757 and assembled, closed and reinforced as specified in the appendix to the box specification. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract, or when there is less than the required amount of the same size per container, mixed sizes may be packed within the same shipping container (see 5.3.1).

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5.2.1.2 Method 2. When specified (see 6.2), three level B shipping containers (600 caps of one type and size only) shall be overpacked in a fiberboard shipping container conforming to class weather resistant, size No. 4 of MIL-B-17757. Toward the end of the contract, or when there is less than the required amount of the same size per container, mixed sizes may be packed within the same shipping container (see 5.3.1).

5.2.2 Level B. Two hundred (200) caps of one type and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container with liner conforming to class domestic, size No. 3A of MIL-B-17757 and assembled, closed and reinforced as specified in the appendix to the box specification. The fiberboard used in fabricating the box and liner shall be grade 275 and 200, respectively. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Toward the end of the contract, or when there is less than the required amount of the same size per container, mixed sizes may be packed within the same shipping container (see 5.3.1).

5.2.3 Commercial. Caps shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rule and U. S. Postal Service Manual or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Mixed sizes. Each shipping container packed with mixed sizes shall have a white paper label, 5 X 4 inches in size, securely attached to the end and side. The words "MIXED SIZES" shall be plainly printed on the labels. Under these words, the sizes and quantities of each size of caps contained therein shall be legibly printed.

6. NOTES

6.1 Intended use. The caps are intended for wear by personnel of the Marine Corps and Navy.

6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procuring requirements.

(a) Title, number and date of this specification.
(b) Size and type required (see 1.2).
(c) Whether first article is required (see 3.2).
(d) Selection of applicable levels of packaging and packing (see 5.1) and 5.2).
(e) When level A, method 2, packing is required (see 5.2.1.2).

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6.2.2 Data requirements. Data conforming to Data Item Descriptions DI-R-4803, DI-R-4805, DI-T-4901, DI-T-4902, DI-T-4903 and DI-T-4904 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Standard samples and shade samples. For information regarding availability of standard samples and standard shades specified, address inquiry to the procuring activity issuing the invitation for bids.

6.4 First article. First article examination, test and approval shall be specified by the contracting officer (see 3.2).

6.5 Figures. Figures 1 through 4 show general design of the cap and are furnished for information only. In the event of conflict between these figures and the text of this specification, the text of this specification shall govern.

6.6 Marginal notations. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

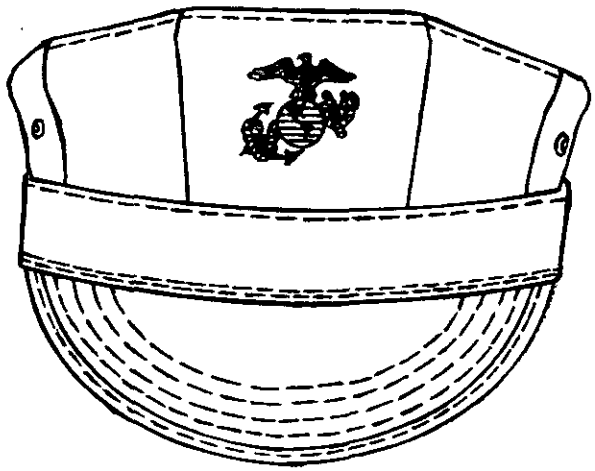
Preparing Activity:
Navy - MC
Project No. 8405-0848

Custodians:
Navy - MC

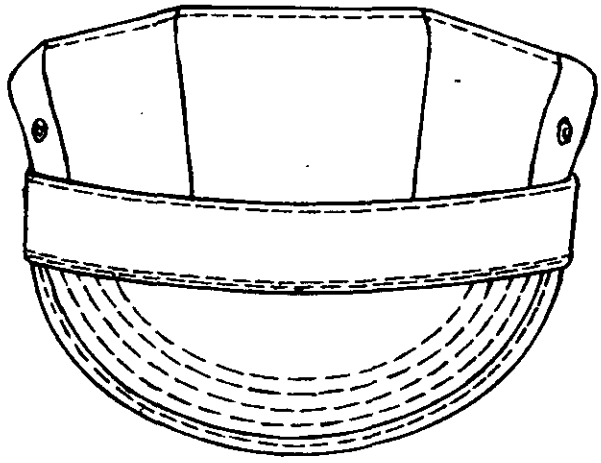
Review Activities:
Navy - NU

User Activities:
Navy - YD

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TYPE I



TYPE II

FIGURE 1. CAP, UTILITY, CAMOUFLAGE.

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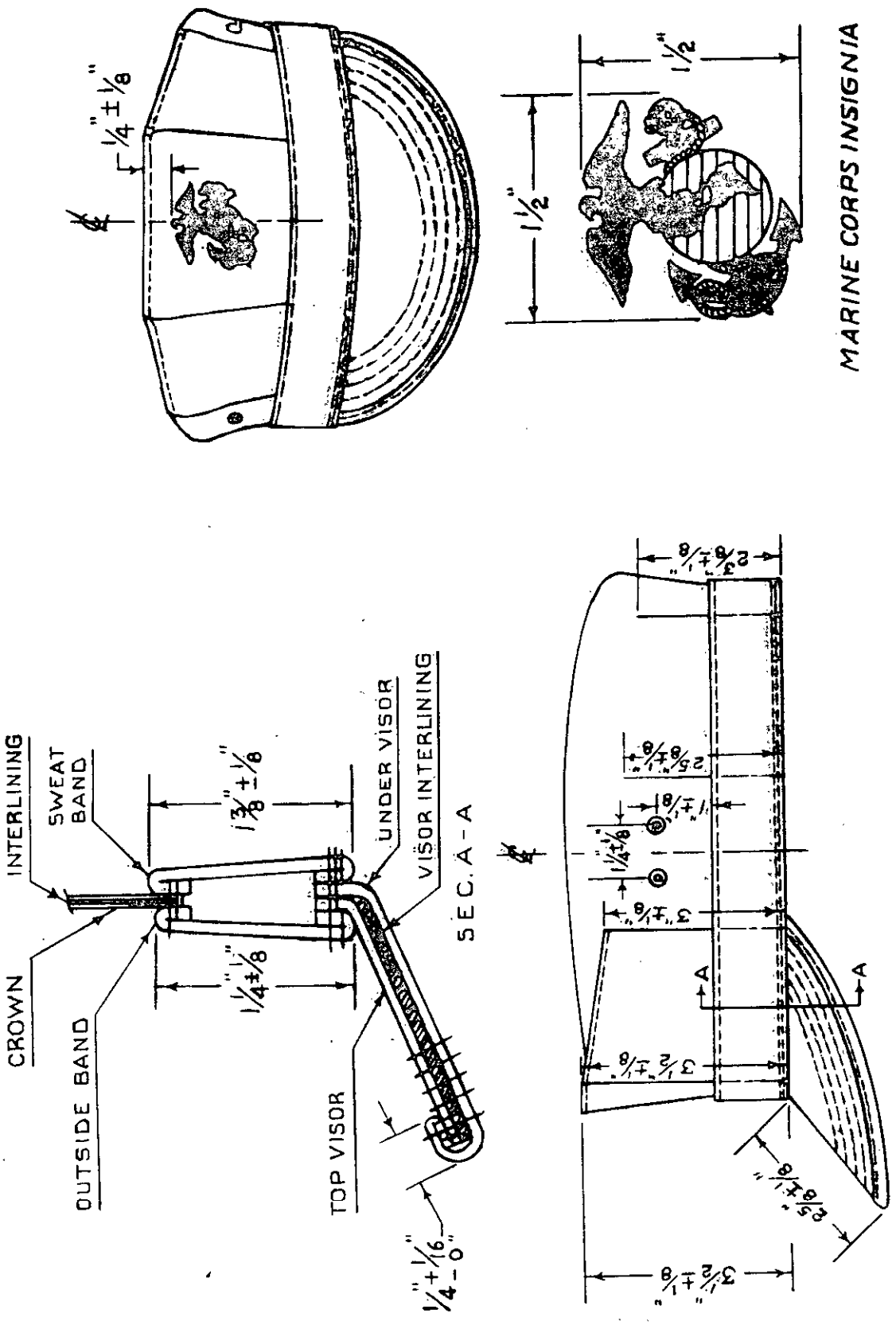


FIGURE 2. CAP, UTILITY, CAMOUFLAGE

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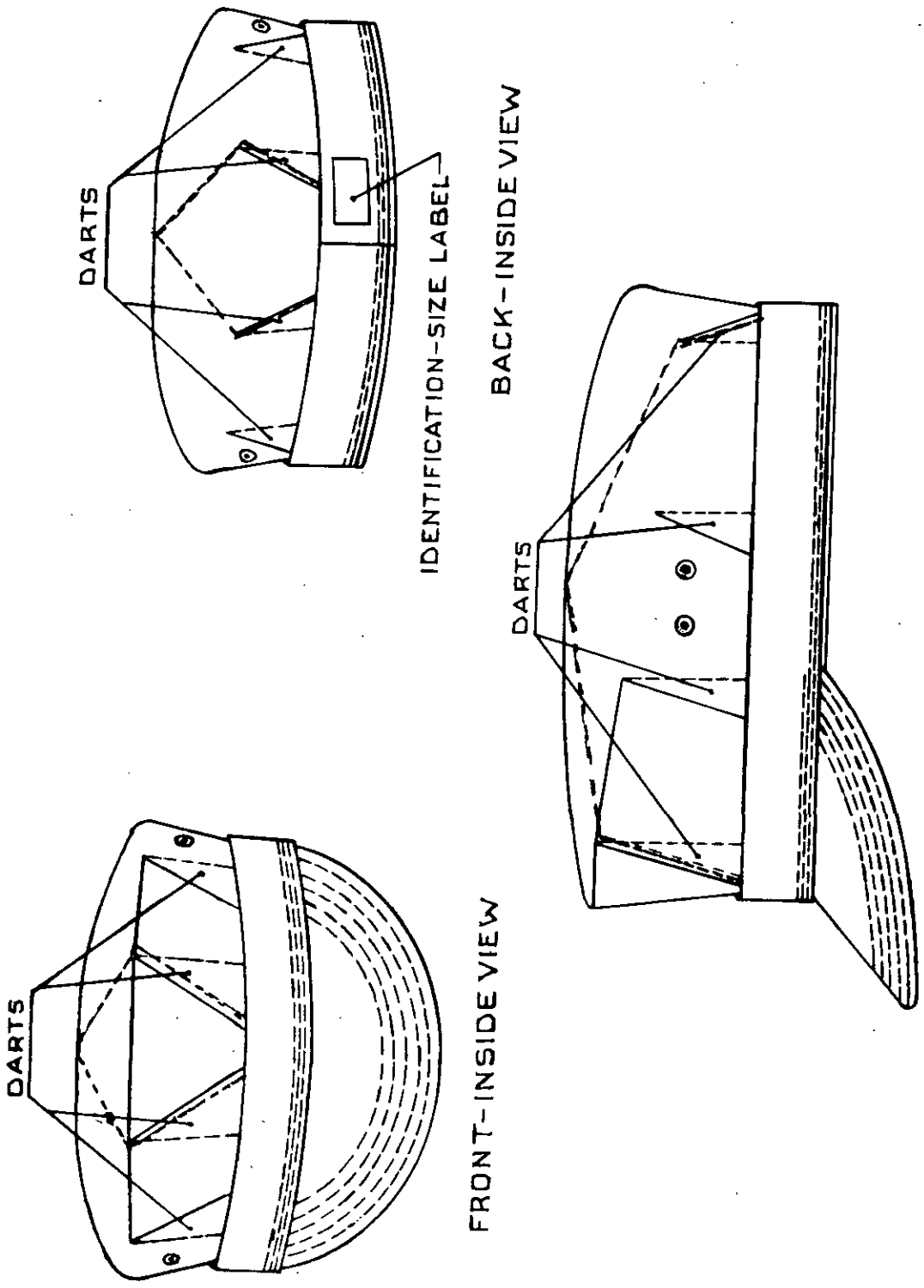


FIGURE 3. CAP, UTILITY, CAMOUFLAGE

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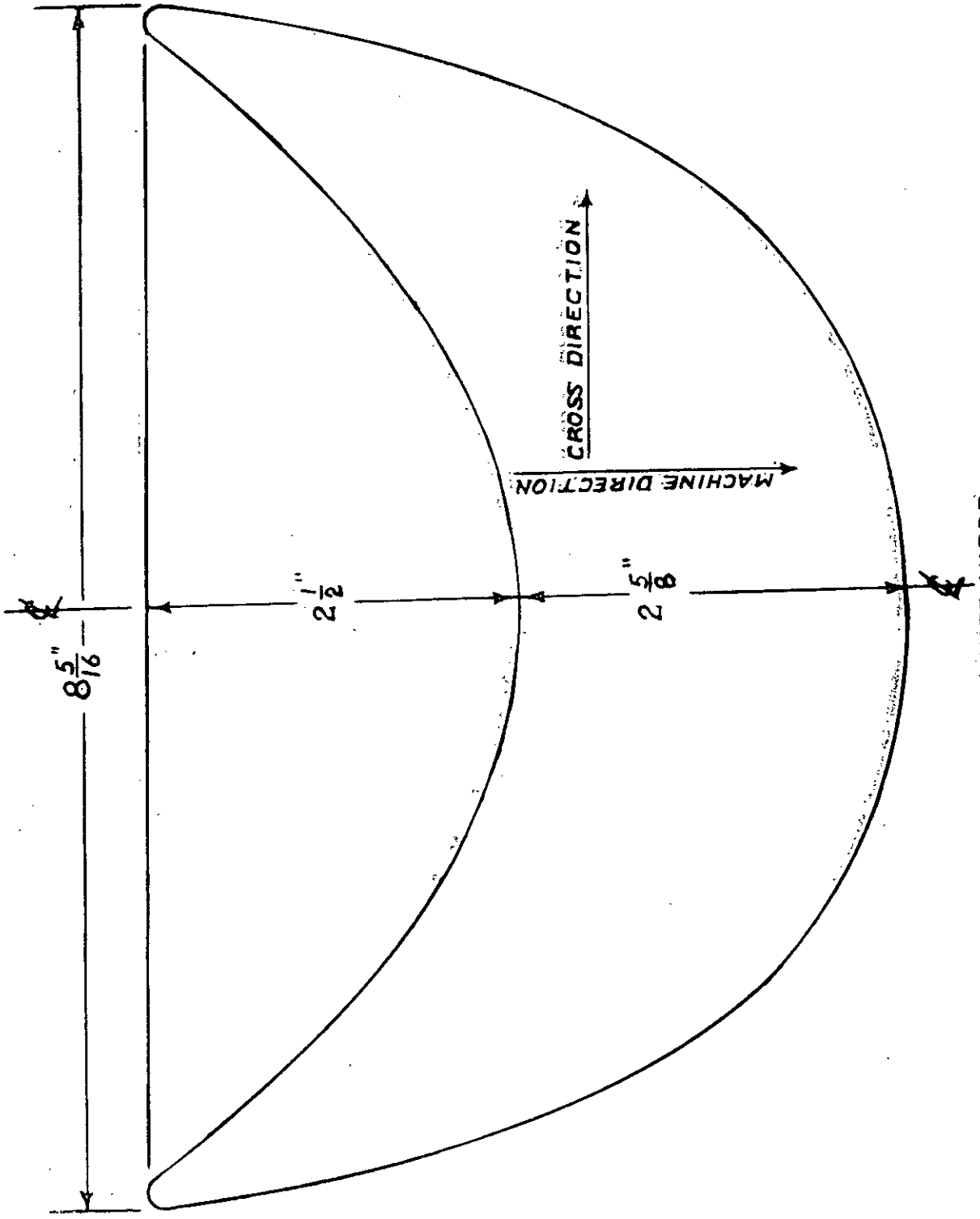


FIGURE 4. CAP, UTILITY, CAMOUFLAGE

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No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-29366A(MC) CAP, UTILITY: CAMOUFLAGE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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